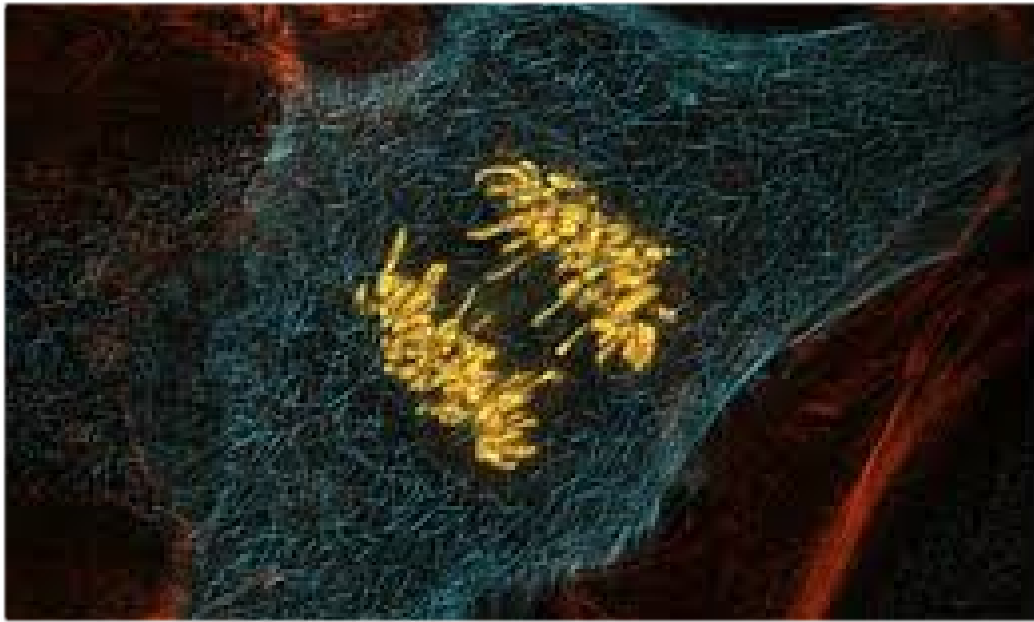


Cellular Replication



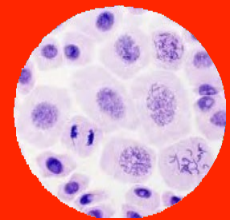
Growth and the Creation of Life

One of the characteristics of living things is the ability to replicate and pass on genetic information to the next generation. Cell division in individual bacteria and archaea usually occurs by binary fission. Mitochondria and chloroplasts also replicate by binary fission, which is evidence of the evolutionary relationship between these organelles and prokaryotes.

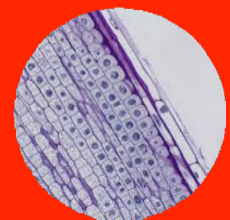
Cell division in eukaryotes is more complex. It requires the cell to manage a complicated process of duplicating the nucleus, other organelles, and multiple linear chromosomes. It is controlled in the cell cycle, which is divided into three parts: interphase, mitosis, and cytokinesis. We split those further for ease of study. Let's start with interphase, which is broken into three stages. In the first growth phase (G₁), the cell grows and prepares to duplicate its DNA. In the synthesis phase (S), the chromosomes are replicated. In the second growth phase (G₂), the cell prepares to divide.



Cellular Cycle and Replication



A step by step guide to growing a human!



Mitosis and Meiosis

Similar processes with VERY different results!